Danmarks Tekniske Universitet

DTU

Final Assignment
Thermo Coupling

Authors of Group 10

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Group Responsibilities

Author	Task
Aikaterini Glynou – s252864	1. Theory
Laurenz Walter Torni – s252851	2. Validation
Laurenz Walter Torni	3. Piston Investigation - Thermal Loads
Aikaterini Glynou – s252864	3. Piston Investigation - Theory
Aikaterini Glynou – s252864	4. Piston Investigation - Simulation
Laurenz Walter Torni – s252851	5. Eigenfrequency Analysis

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1 Theoretical Code Modifications for the Thermal Coupling Analysis

Several aspects were modified in order to generate the Thermal Coupling Problem. Initially, the thermal problem was established, by redefining all the relevant parameters. After deriving the nodal Temperature Distribution, the total structural load was calculated by incorporating both the thermal and the structural loads.

1.1 Thermal Problem In the thermal analysis, each node has only one degree-of-freedom, since the temperature (T) is a scalar field. This dictates the size of the shape function vector and the thermal gradient matrix. Therefore, the following parameters are described in the 'Thermoshapes subroutine':

- Shape Function (N): Is a 1×4 vector, which interpolates the Temperature at every node, to any point inside the element e. So N represents N_1, N_2, N_3, N_4.
- Thermal Gradient Matrix (bmat): A 2×4 matrix. The thermal gradient is $\{\nabla T\} = \begin{Bmatrix} \partial T/\partial x \\ \partial T/\partial y \end{Bmatrix}$. Using the shape functions:

$$\{\nabla T\} = [\mathbf{B}]_T \{T_e\}$$

where $[\mathbf{B}]_T = \begin{bmatrix} \partial N_1/\partial x & \partial N_2/\partial x & \partial N_3/\partial x & \partial N_4/\partial x \\ \partial N_1/\partial y & \partial N_2/\partial y & \partial N_3/\partial y & \partial N_4/\partial y \end{bmatrix}$.

- Thermal Conductivity Matrix (ke): A 4×4 that represents the relevant stiffness matrix of the thermal problem, for each element.

  The matrix is defined by the integral:

$$[\mathbf{k}] = \int_V [\mathbf{B}]_T^T [\kappa] [\mathbf{B}]_T dV, \qquad [\kappa] = k[\mathbf{I}] = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}, \text{for isotropic material}$$

  k: conductivity input **[[1]]**

  The conductivity Matrix is defined in the Fortran script in a new subroutine, where the thermal parameters (Shape functions, Bmat) are required. For its construction, isoparametric formulation is considered.

- The three different thermal loads (surface heat flux, convection, and internal heat generation) were defined in two separate subroutines, while they were combined afterwards to obtain the final temperature distribution. (referencing the first and fourth uploaded images).

  1. **Heat Flux** (rb in $plane42_fluxe$): Load due to prescribed heat flux (f_B) on a surface (S).

$$\{\mathbf{r}_B\} = \int_S [\mathbf{N}]^T f_B dS$$

  The subroutine calculates the load due to heat flux when a second coefficient is not present, since the input file is set up as: SPH, element # , face # ,HFLU, 0 , value.

2. **Convection** (rh in <#C7#>): Load due to convection in a surrounding fluid at T_{fl} and a prescribed convection coefficient, h.

$$\{\mathbf{r}_h\} = \int_S [\mathbf{N}]^T h T_{fl} dS$$

Unlike heat flux, the convection load is calculated when a second coefficient (T_{fl}) is prescribed in the input file, since: SPC, element # , face # , CONV, 0, conv. coef. , bulk temp.

3. Heat Generation (rq in $plane42_rq$): Load due to volumetric heat generation (Q).

$$\{\mathbf{r}_Q\} = \int_V [\mathbf{N}]^T Q dV$$

In the subroutine $plane42_rq$, this is calculated as a volume integral using the Gaussian points ($ng = 2$) : $rq = rq + n * Q * detjac * thk * wi * wj$

The total thermal load (p_therm) is constructed in the fea module, through the tempload subroutine for each case (1, 2, 3), mentioned above.

- **Nodal Temperature Distribution** ($\{T\}$) The nodal temperature distribution for the thermal problem is assembled as:

$$[\mathbf{K}]_T\{\mathbf{T}\} = \{\mathbf{R}\}_T$$

where $[\mathbf{K}]_T$ is the global thermal conductivity matrix (cdmat in buildconductivity) and $\{\mathbf{R}\}_T$ is the global thermal load vector (p_therm)

This temperature distribution is contributing to the total structural load, as described below:

Thermal Equation

$$[\mathbf{K}_{TT}]\{\mathbf{T}\} = \{\mathbf{R}_T\}$$

Structural Equation ($*$)

$$[\mathbf{K}_{DD}]\{\mathbf{D}\} = \{\mathbf{P}_D\} - \underbrace{[\mathbf{K}_{DT}]\{\mathbf{T}\}}_{\text{Coupling Load ge}}$$

- **Thermal-Structural Coupling Vector (ge)** The temperature distribution acts to the structure as a thermal prestrain. Therefore, from equation 3.12 of the book, the vector ge is developed in order to calculate the contribution of the thermal loads to the structural problem.

$$\{\mathbf{ge}\} = \int_V [\mathbf{B}]^T[\mathbf{C}]\{\boldsymbol{\epsilon}\}_T dV \qquad , \quad \{\boldsymbol{\epsilon}\}_T = \{\alpha T\} = \begin{Bmatrix} \alpha T \\ \alpha T \\ 0 \end{Bmatrix}$$

where α = prescribed thermal expansion, $[\mathbf{B}]^T$ = structural B matrix 3×8 and $[\mathbf{C}]$ is the constitutive matrix, 3×3)

- **Total Mechanical Loads**: They are calculated via buildload subroutine by simply applying the Structural Equation ($*$). Therefore, the total load: $p(edof) = p(edof) - ge$

2 Validation of the Thermo Coupling Method

The validation of the thermo coupling implementation was divided into two components, verification of the temperature field distribution and validation of the thermally induced stress field. While the first component ensures an accurate calculation of the thermal distribution, the second validates the thermomechanical coupling through implementation of thermal prestrain effects.

2.1 Validation of the Thermal Distribution

Three distinct thermal loading scenarios were examined to validate the temperature field computation, the heat flux, different temperatures at the boundaries, convective heat transfer, and internal heat generation. An additional combined loading case incorporating all three mechanisms was also considered.

As an initial verification step, Fourier's first law of heat conduction was evaluated for each scenario. The steady-state heat equation in one dimension with a constant heat flux Q is given by

$$-k\frac{dT}{dx} = q, \tag{1}$$

where the thermal conductivity k was chosen to be 1 and 12.5 $\frac{W}{m^2K}$ in different scenarios.

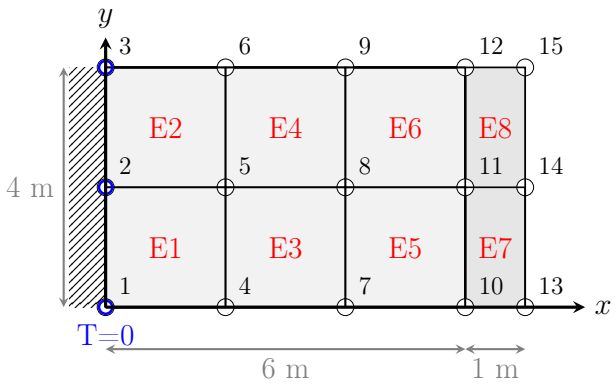


Figure 1: Validation test scenario of a 3 × 4 mesh

Linear temperature profile with prescribed boundary temperatures To validate the temperature distribution computed by the finite element solver, the first test case uses Dirichlet boundary conditions $T(0) = 0\,°\text{C}$ and $T(L) = 100\,°\text{C}$ without any heat generation, convection or heat flux. For steady-state one-dimensional heat conduction with constant thermal conductivity k, the governing equation reduces to

$$\frac{d}{dx}\left(-k\frac{dT}{dx}\right) = 0 \quad \Rightarrow \quad -k\frac{dT}{dx} = q_0 \text{ (constant)}. \tag{2}$$

Integrating twice and applying the boundary conditions yields the linear temperature profile

$$T(x) = \frac{100\,°\text{C}}{L}\,x. \tag{3}$$

Prescribed heat flux $q \neq 0$ For a prescribed heat flux boundary condition, integrating the steady-state heat conduction equation once and applying the boundary condition $T(0) = 0\,°\text{C}$ yields

$$-k\,T(x) = qx + C \quad \Rightarrow \quad T(x) = -\frac{q}{k}\,x. \tag{4}$$

This analytical solution is applicable to all thermal loading scenarios by computing the equivalent heat flux for each boundary face. For convective boundary conditions, the heat flux is determined from the convection coefficient h and the temperature difference ΔT, while for volumetric heat generation, an equivalent surface heat flux is derived from the specific heat generation rate within the cubic elements.

Having established the analytical solutions for validation, the loading magnitudes for each finite element test case must be quantified and compared against the corresponding analytical

predictions. Table 1 summarizes all load cases with their finite element input parameters and equivalent analytical expressions. For convective boundary conditions, the resulting heat flux is given by $\dot{q}_{\text{analytic}} = h(T_{\text{bulk}} - T_{\text{surface}})$, where the surface temperature $T_{\text{surface}} = 0\,^{\circ}\text{C}$.

For the volumetric heat generation case applying heat generation directly in elements E5 and E6 would introduce discrepancies due to the inherent discretization of the domain. To represent volumetric heat generation as an equivalent surface heat flux, two additional elements (E7 and E8) are introduced into the mesh shown in Figure 1. The volumetric heat generation is applied within these elements, resulting in an equivalent heat flux boundary condition on the element face connecting nodes 10 through 12:

$$q_{\text{nodes }10-12} = \frac{\dot{q}_{\text{Heatgeneration,E7+E8}} \cdot V_{\text{E7+E8}}}{A_{\text{face,E5+E6}}} \tag{5}$$

where $\dot{q}_{\text{Heatgeneration,E7+E8}}$ is the volumetric heat generation rate, $V_{\text{E7+E8}}$ is the combined volume of elements E7 and E8, A_{face} is the area of the boundary face, ensuring dimensional consistency.

In Figure 2 the results of the analytical solution and the FEM simulation are presented. As

Table 1: Boundary conditions and heat flux magnitudes of each test case

Scenarios	**Boundary Temperature**	h	T_{bulk}	$\dot{q}_{FEM-input}$	$\dot{q}_{analytic}$
1. Boundary Temp.	$T(0) = 0,\ T(L) = 100$	–	–	–	–
2. Heat flux	$T(0) = 0$	–	–	$80\ \frac{\text{W}}{\text{m}^2}$	$80\ \frac{\text{W}}{\text{m}^2}$
3. Convection	$T(0) = 0$	$0.25\ \frac{\text{W}}{\text{m}^2\text{K}}$	80°C	–	$20\ \frac{\text{W}}{\text{m}^2}$
4. Heat generation	$T(0) = 0$	–	–	$50\ \frac{\text{W}}{\text{m}^3}$	$50\ \frac{\text{W}}{\text{m}^2}$
5. All load cases	$T(0) = 0$	$0.25\ \frac{\text{W}}{\text{m}^2\text{K}}$	80°C	$80\ \frac{\text{W}}{\text{m}^2}$ & $50\ \frac{\text{W}}{\text{m}^3}$	$150\ \frac{\text{W}}{\text{m}^2}$

shown by the datapoints the Code produces the same results as the analytical solutions for every load case as well as combined load cases. To avoid only validation on k and t at 1, a second validation round ffor the same scenarios has been conducted with k = 12.5 and t = 1.3 validtating the numerical temperature distribution calculation also for non uniform parameters.

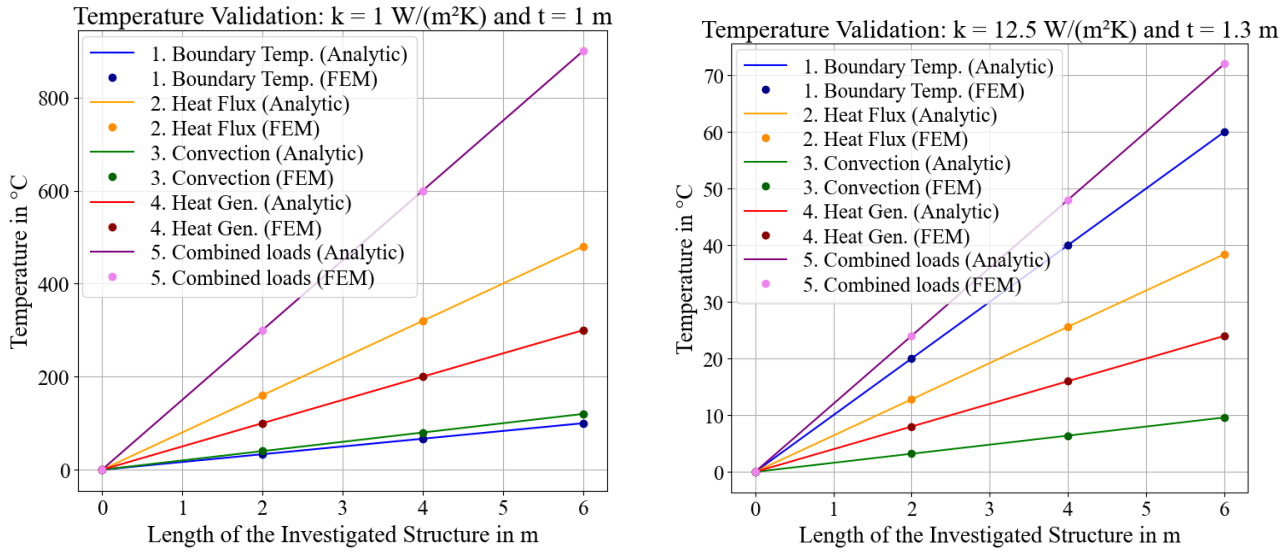


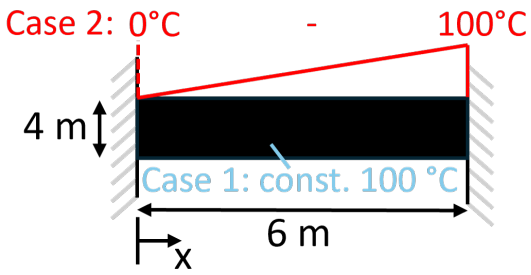
Figure 2: Validation for different Material parameters

2.2 Validation of the Prestress

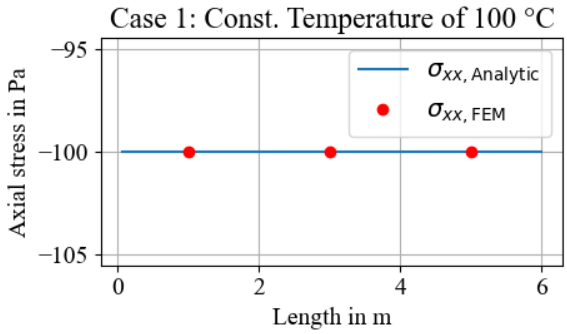
For the validation of the thermocoupling a the stress distribuiton of a beam constrained on both sides as shown in figure 3 was considered. The thermal loading was by setting the whole beam to a constant temperature $T = 100\,\mathrm{K}$ in the second case the beam was given a temperature gradient from the left side at 0 °C to the right side at 100 °C. For a constant temperature the thermal strain is $\epsilon_{thermo} = -\alpha \cdot dT$ which results in a structural stress of

$$\sigma = \epsilon_{\text{mechanical}} + \epsilon_{\text{thermo}} = -\alpha \cdot dT \tag{6}$$

Since the analytical solution is a 1D problem the Poissionsratio has to be set to zero.



(a) A beam fixed on both sides with a youngsmodulus of 1 Pa.



(b) Calculated axial stress due to increased temperature in the bar.

Figure 3: Validation test scenario of a 3 × 4 mesh

Figure 4 compares the analytical axial stress distribution in a fixed–fixed rod under a linear temperature gradient (Figure 3a) with finite element calculations. Without mechanical loads, thermal expansion alone governs the stress state. For a fully constrained rod, the analytical solution yields a linear relationship between axial stress σ and temperature differential ΔT, which the FEM reproduces well across most of the range.

A very good agreement is observed between the analytical curve and the computed FEM stress values for intermediate temperature increments. The deviation remaining between the methods is nearly constant and small, indicating that the model correctly captures the thermoelastic behaviour of the structure.

However, slight discrepancies occur near the boundaries, particularly in the intervals between $x = 0$ m to 1 m and between $x = 5$ m to 6 m. These deviations potenially arise from the numerical treatment

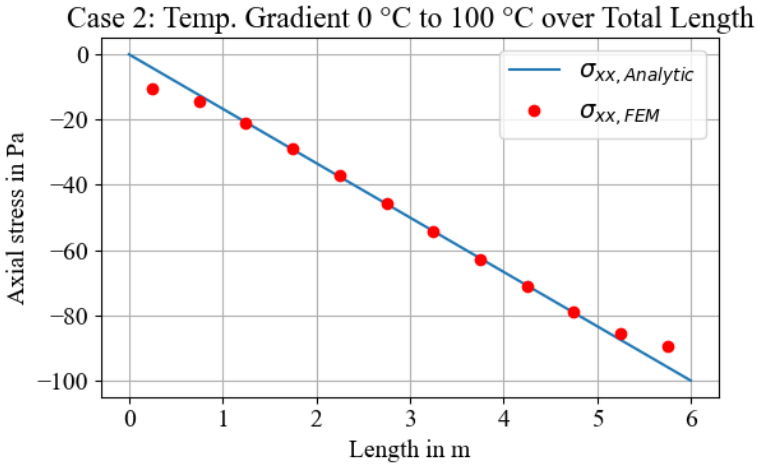


Figure 4: Stress in the bar along x with increasing temperature

of the boundary conditions in the finite element model. Because the rod is represented by elements and constrained at multiple nodes across the cross-section, small local constraint stiffness differences are introduced.

Furthermore, the fixed boundary nodes experience multi-axial constraint conditions, whereas the analytical solution assumes a purely uniaxial stress state. Near the clamped ends, the FEM solution includes localized effects such as transverse constraint and element distortion, which do not appear in the one-dimensional analytical formulation.

Overall, the agreement between the analytical and numerical solutions is accurate. The small deviations at the rod ends are caused by the boundary conditions and reasonable, considering the overall trend of the stress the behaviour modelled by the finite element mehtod seems reasonable.

3 Thermal Coupled Investigation of a Piston

Calculating the load cases In the following the thermal load of a piston is going to be analyzed. The following section presents a concise numerical example used to illustrate the combined thermal and mechanical loads acting on an aluminium piston during a complete four-stroke cycle. All temperatures, heat transfer coefficients and geometric parameters are *assumed* but selected to represent realistic operating conditions of a passenger-car spark-ignition engine.

The primary goal of this example is to quantify (i) the heat input and removal over one cycle, (ii) the resulting mean piston temperature change per cycle, and (iii) representative mechanical forces resulting from gas pressure and inertia.

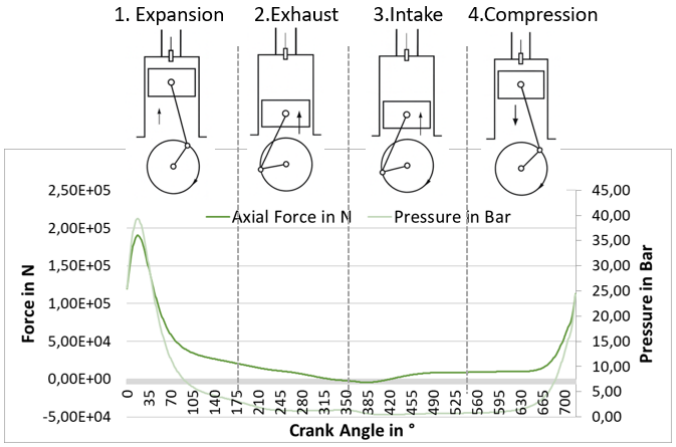


Figure 5: Force and Pressure in a 4 Stroke cycle

Assumptions and Basic Parameters The piston geometry and material constants used in this analysis are summarised in Table 2. These values represent a typical aluminium piston with moderate bore size. The stroke duration corresponds to an engine speed of 5000 rpm, which is used to convert heat fluxes into per-stroke energy transfers.

Table 2: Engine geometry and component parameters.

Parameter	Value
Cylinder diameter D	80 mm
Gas overpressure in cylinder p_u	2 MPa
Piston pin diameter d	50 mm
Piston pin bearing width b	25 mm
Engine speed n	5000 rpm

The assumed gas temperatures and thermal coefficients for each stroke are given in Table 3, taking into consideration the maximum gas pressure is equal to 30.9 bar. The resulting transfer heat flux is calculated by using following equation.

$$q_k = h_k \ (T_{g,k} - T_p)$$

The resulting energies for each stroke are summarized in Table 3.

Table 3: Assumed temperatures and heat transfer coefficients for a typical engine.

Stroke	Gas Temp. T_g	Conv. Coefficient h_g	Piston Temp. T_p	q_{transfer}
1. Expansion	1900 C	3000 $\mathrm{W\,m^{-2}\,K^{-1}}$	420 C	4.44 $[\mathrm{MW\,m^{-2}}]$
2. Exhaust	850 C	400 $\mathrm{W\,m^{-2}\,K^{-1}}$	420 C	172 $[\mathrm{kW\,m^{-2}}]$
3. Intake	320 C	100 $\mathrm{W\,m^{-2}\,K^{-1}}$	380 C	-6 $[\mathrm{kW\,m^{-2}}]$
4. Compression	800 C	800 $\mathrm{W\,m^{-2}\,K^{-1}}$	400 C	320 $[\mathrm{kW\,m^{-2}}]$

Mechanical Load Estimation

For the particular study case we examined the piston at the position after the TDC and right after the ignition (in our case 18° crank angle). At that point, maximum gas pressure is reached in the ignition chamber and also maximum total axial force is achieved at the piston. The piston total axial force is the sum of forces from the combustion gases and the inertia forces acting on the piston by the oscillating components connected to it. The oscillating components of an engine are the engine pistons and part of the connecting rod (usually two–thirds of the mass closest to the gudgeon pin). Therefore, the two main mechanical loads acting on the piston were calculated by the following formulas.

$$F_g = \frac{D^2\,\pi\,P_{\text{cyl}}}{4} \qquad F_{os} = (m_k + m_{pl2})\,r_{\text{crank}}\,\omega^2\,(\cos\alpha + \lambda\cos 2\alpha) \tag{6}$$

where

D = Cylinder bore — P_{cyl} = Cylinder gas pressure

m_k = Piston mass — m_{pl2} = Oscillating mass of connecting rod

r_{crank} = Crank radius — ω = Rotational speed

α = Crank angle — $\lambda = \dfrac{l_{pl}}{r_{\text{crank}}}$ (rod/crank ratio)

The total axial force, therefore, can be calculated by the following equation. The mechanical loads are presented in Table 4:

$$F_T = F_g + F_{os} \tag{8}$$

Table 4: Mechanical loads acting on the piston

Load Type	**Value**
Peak Gas Force F_g	−19.8 kN
Inertia Force F_{os}	8.3 kN
Total Axial Force acting on the piston	−18.9 kN

3.1 Model Set-up

The geometry for this study was designed in Solidworks, as a surface, in compliance to the data presented in Table 2, while the crank shaft was equal to 26mm and is illustrated in Figure 6. In order to create an input file fo Fortran, ANSYS was used to mesh the structure, set the material,the boundary conditions and the loads. After extracting the final file from ANSYS, manual modifications were also necessary to adjast the temperature boundaries and the heat flux.

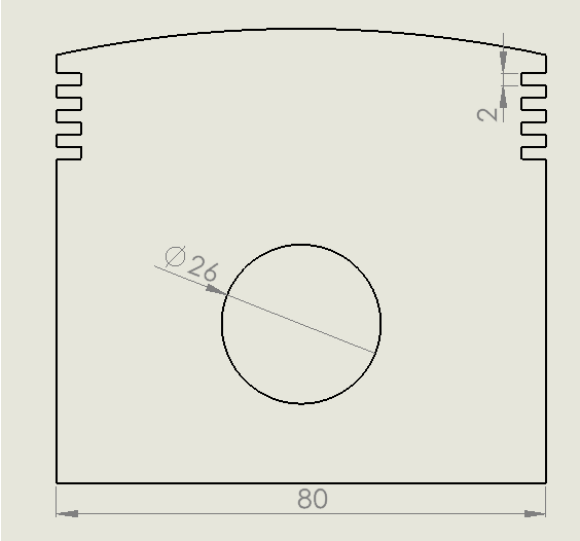


Figure 6: 2D Piston sketch

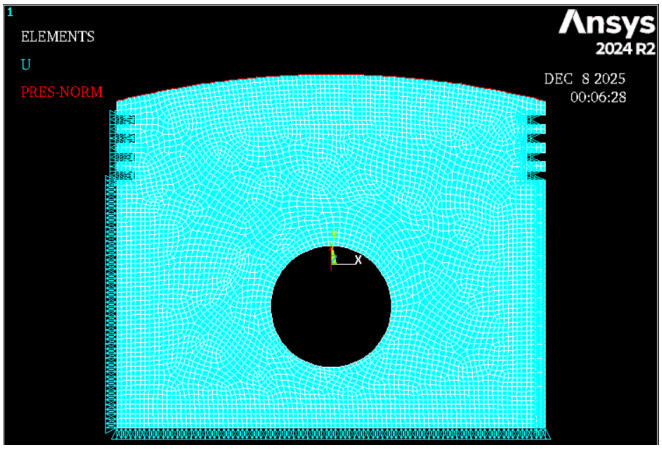


Figure 7: Set-Up Model with loads and mesh

Material Properties The material properties have been set-up for the simulation according to the Table 5, assuming the piston's material to be an aluminum alloy (Figure 8)

Table 5: Material Properties

Property Name	**Processor inp**	**Symbol**	**Value**
Young's Modulus	EX	E	5.5×10^{10} Pa
Thermal Expansion Coefficient	ALPX	α	2.4×10^{-6} K^{-1}
Density	DENS	ρ	2700 kg/m^3
Thermal Conductivity	KXX	k	200 $W/(m \cdot K)$
Specific Heat (Capacity)	C	c	864 $J/(kg \cdot K)$
Poisson's Ratio	PRXY	ν	0.33
Thickness	R	t	2.0×10^{-2} m

Type of Material	Young's modulus (MPa)	Poisson's ratio	Density (kg/m^3)	Thermal conductivity (W/m.K)	Thermal expansion coefficient (10^{-6} K^{-1})
Austenitic steel	190000	0.33	7800	20	12
Steel-Cr-Mo	190000	0.33	7800	40	13
Aluminum alloy	55000	0.33	2700	150	24
AS12 UNG fibre-reinforced	55000	0.33	2700	100	20
Cast iron GS	160000	0.33	7200	35	35

Piston material properties in SI units [20].

Figure 8: Standard pistons materials [2]

Meshing In this simulation, Solid 182, 4-node quadrilateral elements are utilized and the Element Behavior was set up to: Plane Strain with Thickness (plane strs w/thk).

The meshing could be considered fine, since the total number of elements used in this model is 5502. (Figure 7)

Finally, the mesh size was equal to 1, in order to achieve a fine mesh, but maintain good computational efficiency.

Boundary Conditions

- **Roller Slider Fixtures:** These were applied at the bottom of the piston ($UY = 0$) as well as on the left and right edges ($UX = 0$), where the piston is constrained by the cylinder.
- **Thermal Boundaries:** Thermal boundary conditions were established to simulate the oil temperature in contact with the piston. The outer areas of the piston, except for the crown, were assigned a constant temperature of 100°C, while the upper two rings, which do not come into contact with the engine oil, were assigned a constant temperature of 120°C.

Load Application In this particular study, the load case concerns both mechanical and thermal loads for the simulation of the piston right after the ignition point, when the forces reach ther maximum magnitude.

The mechanical forces (calculated in Table 4) were applied after converting them to equivalent pressure loads (surface loads) (**P**)

Table 6: Conversion of Applied Forces (**F**) to Equivalent Pressure Loads (**P**)

Load Source	**Applied Pressure**	**Numerical Value**
Combustion Force (F_g)	$P_{\text{crown}} = F_g / A_{\text{crown}}$	$19,800 \text{ N} / (\pi \cdot (0.08 \text{ m})^2/4) \approx \mathbf{3.94}$ MPa
Inertia Force (F_c)	$P_{\text{pin}} = F_c / A_{\text{bearing}}$	$8,300 \text{ N} / (0.08 \text{ m} \cdot 0.026 \text{ m}) \approx \mathbf{3.99}$ MPa

The Combustion Gas Force and the Inertia Force from the oscillating parts were applied at the combustion crown and piston pin boss surface, respectively.

Furthermore, heat flux was applied to the piston crown to simulate the heat transfer from the combustion gases. The magnitude used was **4.44** MW/m$^{\mathbf{2}}$ (equivalent to $4,440,000$ W/m^2 in the input file).

To simulate the cooling effect of the engine oil on the piston, two types of boundary conditions were used, as mentioned above.

Results The Nodal Temperature Distribution is presented in Figure 9, meanwhile the Von Mises stresses are indicated in Figure 10.

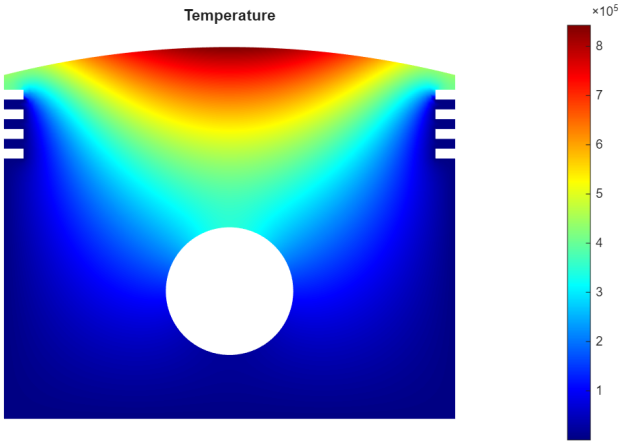


Figure 9: Nodal Temperature Distribution

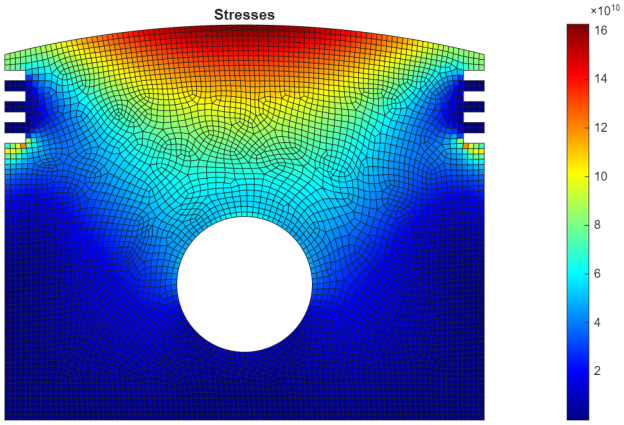


Figure 10: Von Mises Plot

It is obvious that the highest temperatures are concentrated at the piston crown (the top surface). This is expected because the crown is where the extremely high heat flux ($\mathbf{q}$) from combustion is applied. Moreover, the heat flows primarily downward and outward from the crown toward the cooling boundaries, which is a good representation of a real life case. However, the temperatures are extremely at the top of the piston, possibly because In a real piston, heat flows around the piston and across the component. A 2D cross-section limits the number of paths available for heat to flow away from the hot crown to the cooler surfaces and therefore leading to artificially internal temperatures.

In the Von Mises Plot, it is clear that these extreme temperatures have affected significantly the stresses. The most prominent stresses are located at the crown of the structure, due to the heat flux, as well as the pressure of the gas. In addition, some stresses are also visible at the upper area of the piston's pin boss, because of the Inertia Forces applied there.

In addition, the stress concentration at the near end of the last cooling ring is developed due to the boundary condition at the edges of the piston that restricts the thermal expantion and relative displacement. The high stresses in this region might have also occurred due to the sharp geometry of the corner.The stress distribution predicted by the model, however, matches the real-world behavior, even though absolute magnitudes may differ due to model simplifications.

Finally, it is important to mention that a 2D plane strain model restricts heat conduction to only two directions. In reality, heat can flow in three dimensions, including circumferentially around the piston. This additional heat path in a real piston would reduce the peak temperatures and, as a result the total stresses.

Thus, the magnitudes of the stresses and the temperature might not meet the expected and realistic value, nevertheless, the simulation highlights the regions most susceptible to thermal fatigue or structural failure.

4 Eigenfrequency Analysis

Figure 11 shows the modal behaviour of a 20m fixed–fixed steel beam at various temperatures. The left plot depicts angular frequencies ω (in rad/s), and the right plot the eigenvalues $\lambda = \omega^2$ on a symmetric logarithmic scale. With increasing temperature, several low-order

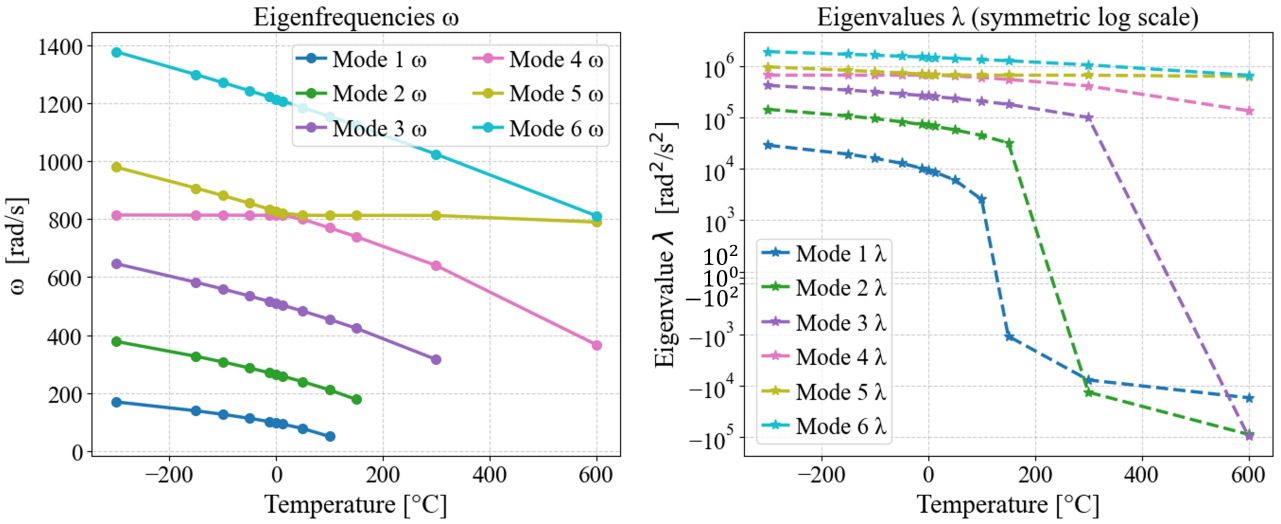


Figure 11: Eigenfrequencies of a fixed-fixed steel bar (L=20 m) at various temperature levels. Material: E=210 GPa, α=12×10^{-6} K^{-1}, ρ=7850 kg/m^3, ν=0.3.

eigenvalues λ decrease and eventually become negative. The first mode becomes negative at 150°C the second at 300°C, and the third at approximately 600°C, indicating a mode-dependent thermal behaviour. Once $\lambda < 0$, the corresponding frequency curve terminates since $\omega = \sqrt{\lambda}$ is undefined. In general, the eigenfrequencies increase with decreasing temperature due to tensile stresses induced by constrained contraction, whereas they decrease with rising temperature. An exception is Mode 5, which initially almost coincides with Mode 4 but deviates from the overall decreasing trend at higher temperatures. In general the trend of λ trend reflects thermal-induced axial compression, reducing bending stiffness under clamped ends. Furthermore, it is observed that the eigenvalues do not diverge to negative infinity but instead reach magnitudes comparable to their positive counterparts, differing only that they are negative. This eventually shows that the system retains structural characteristics under high thermal loads. However, when plotting the modeshapes (Figure 12), it is observed that the modeshape evolves from half a sinusoidal wave to a full sinusoidal form when λ becomes negative at 300°C, and to one and a half sinusoidal waves at around 600°C.

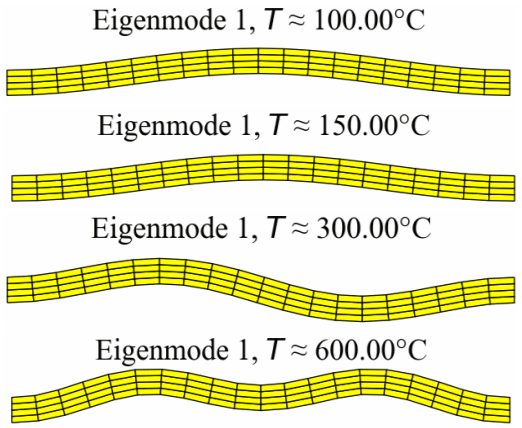


Figure 12: Eigenmode 1 shapes at different temperatures.

Validation attempts and outcome Validation was attempted by comparison with analytical Euler–Bernoulli eigenfrequency solutions, but exact agreement could not be achieved. Nevertheless, the eigenvalue computation and thermal prestrain formulation have been validated in prior studies. The remaining uncertainty lies in the formulation of the stress stiffness matrix and its coupling to the global stiffness matrix.

References

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